Gilberto Romero-Cano

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SUMMARY

Computer Science student at CU Denver (May 2027). I build GPU and ML systems and then check whether they actually work: validating against published results, scoring against simple baselines, and reporting what didn't pan out. Looking for ML engineering internships and new-grad roles.

EDUCATION

University of Colorado Denver · B.S. in Computer Science — Expected May 2027

Coursework: Algorithms & Data Structures, Machine Learning, Software Engineering, Networks, Operating Systems, Databases
NVIDIA DLI certificates (2026): Building RAG Agents with LLMs · Generative AI with Diffusion Models · Transformer-Based NLP

PROJECTS

Raymarcher: GPU Ray Tracer for Curved Spacetime | CUDA C++, WebGPU/WGSL, TypeScript

- Built a compiler that takes a spacetime written in a small metric language, differentiates it symbolically, and generates ray-tracing code for a native CUDA renderer and a WebGPU browser playground
- Checked the physics against published results (Bardeen 1973; Bardeen, Press & Teukolsky 1972) to within 10^{-10}; the CUDA and WebGPU renderers agree on every ray across 14M test pixels
- Profiled the Kerr kernel in Nsight Compute and made it 1.6× faster (fused generated code, register cap for occupancy); it runs at 10.4 TFLOP/s, 53% of fp32 peak on an RTX 3070

predictive-audit: Does Your Forecast Beat Persistence? | Python, time series, bootstrap

- Grew out of my 4-person team's SDN project (Kafka, LSTM forecaster, DQN agent, ONNX INT8 export 3.2× faster), whose F1 ≈ 0.99 matched “next step = this step” (F1 0.993) once I checked
- CLI and library that compares a forecast with persistence, moving-average and seasonal-naive baselines (moving-block bootstrap intervals) and flags shuffled splits and leaky features
- Ran it on 5 published results: 4 don't clearly beat trivial baselines, including an IEEE ICC 2021 paper. One holds up, and the tool says so

F1 Winner Oracle | Python, GitHub Actions, JavaScript

- Calibrated race-winner odds, frozen before each race in a hash-chained log and scored in public. Started from a classmate's fork; the model, evaluation and site are mine
- 174 walk-forward races: log loss 1.24 vs 1.53 for “pole sitter wins” (95% CI excludes 0). Also reports what didn't help, including my earlier XGBRanker

Spatial Pipeline: Wildlife Photo Hotspot Engine | Python, spatial statistics, deck.gl

- Finds where and when to photograph wildlife in iNaturalist and YFCC100M data: equal-area hexes, unique-photographer counts, Moran's I, Getis-Ord Gi* with FDR control, seasonal lift
- On a month of Denver data (6,417 observations, 954 observers), significant hotspots land where a naturalist would point you; Gi* and FDR are tested against reference implementations

Humanoid Locomotion with Deep RL | Python, MuJoCo, PPO

- Trained a 3D humanoid with PPO for 160M steps to walk over uneven terrain, using a 5-point rangefinder so it couldn't memorize the track
- Fixed failure modes with reward shaping (freezing, walking sideways, diving to end episodes early). The agent found a knee-slide down steep ramps that it couldn't unlearn without losing its gait

SKILLS

Languages	Python, C++, CUDA C++, TypeScript, JavaScript, C, C#, Swift, SQL
ML & evaluation	PyTorch, XGBoost, scikit-learn, Stable-Baselines3, ONNX Runtime, walk-forward evaluation, bootstrap intervals
GPU & data	CUDA, NVRTC, WebGPU, Nsight Compute, RAPIDS cuML, pandas, NumPy, SciPy, PyArrow, Dask, Kafka
Tools	Git, GitHub Actions, Docker, Linux, CMake, FastAPI, Streamlit, MuJoCo, Mininet

WORK EXPERIENCE

The Home Depot · Service Desk Associate · Denver, CO — May 2022 – Present

- Resolve customer issues with inventory, installations and order fulfillment in a high-volume store, and work with floor teams to keep inventory records accurate